

ON CLOUD NINE Shubhanshu aboard the International Space Station, Jul. 5, 2025

When Group Captain **Shubhanshu Shukla lifted off aboard Axiom-4** on June 25, 2025—becoming only the second Indian to travel to space—he radioed back a line that captured a long-deferred national ambition: "We're back in space after 41 years.... This is the beginning of India's human spaceflight." A day later, Shukla became the first Indian to enter the International Space Station, settling in for 18 days of living and working 400 kilometres above Earth. It was a personal milestone—but also the most visible expression of a transformation that had been reshaping India's space sector.

That shift had gathered decisive momentum in June 2020, when the Narendra Modi government dismantled ISRO's monopoly over building and launching rockets and satellites. Private players were allowed entry, ISRO facilities were opened to them on payment, and the sector was liberalised to permit 100 per cent foreign direct investment. IN-SPACe was created as a single-window regulator, formalising ISRO's transition from sole operator to ecosystem enabler. The effects were swift. Nearly 200 space start-ups emerged, spanning small satellites, launch vehicles, propulsion, earth-observation, debris tracking and downstream analytics.

Alongside, ISRO anchored India's credibility with a steady run of mission benchmarks. NavIC, India's GPS-equivalent, had become operational by 2018, underpinning defence readiness, disaster response and civilian navigation. Chandrayaan-2 (2019) deepened lunar science despite its lander setback; Chandrayaan-3 (2023) made history by soft-landing the Vikram module near the Moon's south pole. Aditya-L1 (2023) placed India among solar-observing nations. The PSLV remained a reliable workhorse, while the LVM3 carried India's heaviest payloads yet, including OneWeb broadband satellites, signalling ISRO's rise in the commercial launch market.

In parallel, Gaganyaan—India's indigenous human spaceflight programme sanctioned in 2018 and now targeted for 2027—quietly gathered pace. Shukla's flight, a global collaboration, sharpens that effort and feeds a longer arc: a Bharatiya Antariksh Station by 2035 and a crewed lunar mission by 2040. ■

GROUP CAPT. SHUBHANSHU SHUKLA, ***Astronaut***

"I tell the kids one of you may be the one to set foot on the Moon when we eventually go there in 2040, but I will be fighting you for that part"



◀ The Moon Warriors
Sept. 11, 2023

"Somanath succeeded Sivan as ISRO chairman in January 2022... introduced two major changes in the management style. The first, he termed a failure-oriented approach, a variation of the old Murphy's Law according to which everything that can go wrong will go wrong. The second was, as he told his team, "We will not go for the launch of Chandrayaan-3 unless there is 100 per cent satisfaction of all the teams involved." All the niggles thus sorted, Somanath gave the clearance to launch the spacecraft."